

DevOps Lab Programs

Program-4: Integrate Git with Docker for Source-Controlled application Builds

Project Structure:

Program-4/

→Dockerfile

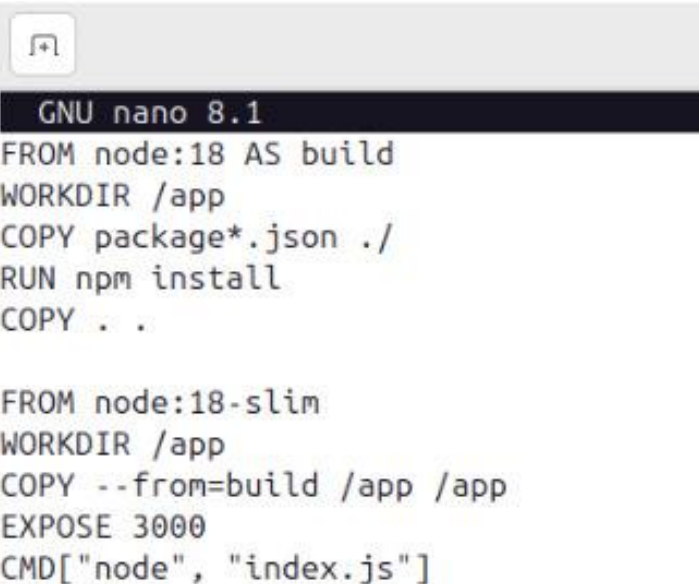
→index.js

→package.json

→package-lock.json

Steps for execution:

Step1: Create a Dockerfile without any extension and add the following content



```
GNU nano 8.1
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step 2: Create an index.js file with following code

```
GNU nano 8.1
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req,res) =>{
res.send("Hello from git+docker integration");
});

app.listen(PORT, ()=>{
console.log(`running on http://localhost:${PORT}`);
});
```

Step 3: Create file package.json by executing the command

`npm init -y`

Step 4: Initialize Git Repository locally.

`git config --global user.name ""`

`git config --global user.email ""`

`git init`

`git add .`

`git commit -m "Initial commit"`

```
Nov 19 16:14
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program5
create node 100644 node_modules/statuses/package.json
create node 100644 node_modules/toidentifier/HISTORY.md
create node 100644 node_modules/toidentifier/LICENSE
create node 100644 node_modules/toidentifier/README.md
create node 100644 node_modules/toidentifier/index.js
create node 100644 node_modules/toidentifier/package.json
create node 100644 node_modules/type-is/HISTORY.md
create node 100644 node_modules/type-is/LICENSE
create node 100644 node_modules/type-is/README.md
create node 100644 node_modules/type-is/index.js
create node 100644 node_modules/type-is/package.json
create node 100644 node_modules/unpipe/HISTORY.md
create node 100644 node_modules/unpipe/LICENSE
create node 100644 node_modules/unpipe/README.md
create node 100644 node_modules/unpipe/index.js
create node 100644 node_modules/unpipe/package.json
create node 100644 node_modules/vary/HISTORY.md
create node 100644 node_modules/vary/LICENSE
create node 100644 node_modules/vary/README.md
create node 100644 node_modules/vary/index.js
create node 100644 node_modules/vary/package.json
create node 100644 node_modules/wrapappy/LICENSE
create node 100644 node_modules/wrapappy/README.md
create node 100644 node_modules/wrapappy/package.json
create node 100644 node_modules/wrapappy/wrapappy.js
create node 100644 package-lock.json
create node 100644 package.json
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4$ git branch -M main
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4$ git remote add origin https://github.com/RakshithaDas/Program4.git
Username for 'https://github.com': RakshithaDas
Password for 'https://RakshithaDas@github.com':
Enumerating objects: 716, done.
Counting objects: 100% (716/716), done.
Delta compression using up to 8 threads
Compressing objects: 100% (661/661), done.
Writing objects: 100% (716/716), 723.12 KiB | 7.02 MiB/s, done.
Total 716 (delta 140), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (140/140), done.
To https://github.com/RakshithaDas/Program4.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4$
```

Step 5: Link to remote repository

Step 6: Check the gitHub repository.

